

SPX IV Surface Forecasting — Results So Far

Setup

Forecasting the S&P 500 implied-volatility surface — **170 cells per day** (10 maturities $\times$ 17 call-equivalent deltas), **21 days** of context, **63 days** ahead. Canonical 70/10/20 train/val/test split.

Test period is **2017-10** $\rightarrow$ **2019-10** (491 windows), which spans Volmageddon (Feb 2018) and the Q4 2018 selloff. All models trained with **MSE in scaled space** for fair comparison.

Headline scoreboard

Best variant of each model.

Model	MSE	MAE	IC mean	t+63 MSE
DynGWN (nh16)	0.169	0.298	0.842	0.240
HOT (product, no-norm)	0.169	0.322	0.841	0.213
DLinear	0.170	0.318	0.913	0.256
HOT (sum, no-norm)	0.176	0.324	0.871	0.248
VAR(BIC) p=1	0.180	0.324	0.849	0.236
PatchTST (no-RevIN)	0.191	0.327	0.897	0.315
Persistence (baseline)	0.223	0.334	0.887	0.388

The top three (DynGWN, HOT-product, DLinear) are within **1%** of each other on overall MSE — but each wins a *different* metric.

What’s statistically defensible

Diebold-Mariano vs Persistence with HAC variance and Holm-Bonferroni correction. The picture below is the one that survives multiple-comparison adjustment with the available 491 windows.

1. DynGWN is the only model that significantly beats persistence overall

Model	DM stat	Holm p	Significant?
DynGWN	3.42	0.005	yes (p<0.01)
HOT (sum)	2.28	0.157	—
HOT (product)	2.20	0.165	—
DLinear	2.17	0.165	—

Model	DM stat	Holm p	Significant?
VAR(BIC)	1.75	0.239	—

Mean improvements over persistence are similar across the top group ($\approx +0.05$). What lets DynGWN cross the bar is **lower variance of the win**: its margin over persistence is more consistent day-to-day. Graph message-passing produces forecasts whose error differential against persistence has lower autocovariance.

2. Five models significantly beat persistence at t+63

Rank	Model	DM stat	Holm p
1	DynGWN	6.05	$< 10^{-7}$
2	HOT (product)	5.20	$< 10^{-5}$
3	VAR(BIC)	4.63	$< 10^{-4}$
4	HOT (sum)	4.30	$< 10^{-4}$
5	DLinear	3.57	0.001

Long-horizon mean reversion is real and reproducible across very different architectures. **DynGWN** has the strongest signal here too.

3. Persistence is the right baseline at t+1

Every non-persistence model is significantly *worse* at t+1. Only DLinear is statistically tied. Worth caveating in the writeup: not every architecture should be asked to beat persistence at every horizon.

Per-model takeaways

Each model has emerged with a distinct character — these are the single-sentence headlines worth quoting:

DynGWN — *Lowest MAE, lowest median MSE, biggest gap on trimmed MSE.* The most precise per-cell predictor on a typical day. Significantly better than persistence overall (Holm p = 0.005); strongest long-horizon DM stat.

HOT (product) — *Regime-shock specialist.* Tightest tail of any model (max per-window MSE 0.64 vs persistence’s 2.12). On the worst day in the test set (Dec 26, 2018, post-Volmageddon recovery), HOT-product MSE = 0.23 vs every other model > 0.7 . Wins absolute t+63 MSE. Pays for it with collapse at t+1 and t+5.

DLinear — *Highest IC, balanced everywhere.* Best at preserving cross-sectional ranking across all horizons (IC 0.913). Doesn’t dominate any single metric but is rarely worst either. Trains in 30 seconds on MPS.

VAR(BIC) — *Closed-form, competitive at long horizons.* BIC formally selects p=1 (AIC overfits to p=6 with 30% worse t+63). Significantly beats persistence at t+63. Two seconds to fit, no GPU needed.

PatchTST — *RevIN strips the level signal.* With per-window normalization on (legacy default), bias = -0.13 and worst long-horizon MSE in the field. Disabling RevIN fixes the bias but PatchTST still doesn’t reach the top tier — the seq_len = 21 / K = 170 / T = 1854-windows regime is too small for transformer attention to add value over channel-independent linear maps.

Robustness

~10% of days drive most of the spread. When the worst-decile windows are removed (ranked by median MSE across models), DynGWN’s lead opens up:

Model	full mean	common-trim 10%
DynGWN	0.169	0.126
DLinear	0.170	0.137
HOT (product)	0.169	0.144
Persistence	0.223	0.169

DynGWN’s headline-MSE lead of 0.6% over DLinear becomes an **8% lead** when outlier days are stripped.

The worst common days cluster in Q4 2018. Eight of the top-10 worst-MSE windows are October 2018; the others are post-Christmas Dec 26 / Dec 31, during the Volmageddon-recovery transition.

Per-horizon claims are seed-sensitive. Tested on DynGWN (nh32): per-window correlation across seeds is 0.96 (predictions are consistent), but per-horizon mean MSE swings 14–29% between seeds. **Aggregate metrics (overall MSE, MAE) are stable across seeds; specific per-horizon numbers should be reported with awareness of this variance.**

Caveats

- **Effective sample size « 491.** Test windows share 62 of 63 forecast days with their neighbours, so DM tests need HAC variance with bandwidth equal to the forecast horizon. With this correction, only DynGWN crosses Holm-significance overall — even though three other models have positive mean differentials of similar magnitude.
- **Ranking among the top three (DynGWN, HOT-product, DLinear) is not statistically distinguishable.** All within 1% on headline MSE, all fail the pairwise DM test against each other under Holm. The substantive findings are the *per-model character* claims, not the leaderboard order.

Next steps

Three things would strengthen the conclusions, in order of expected impact:

1. **Train on the full dataset** (~838 test windows including COVID and the 2022 vol regime). The DM-significance gap is N-limited; full dataset would likely push DLinear and HOT across the bar too.
2. **Seed-sensitivity check on the shrunk DynGWN.** Confirms the headline significance claim doesn’t disappear with a different initialization.

3. **HOT retune at uniform compute budget.** Current HOT (product, no-norm) trained for ~ 50 epochs vs DynGWN's 64 and DLinear's 31 — a controlled comparison would let us claim “best of each architecture” cleanly.